

BBT2202/CNS2202: Computer Organization & Architecture

Lecture 1:

Introduction to Computer Organization & Architecture

Dr. Victor Mageto

School of Computer & Engineering Science

Strathmore University

Outline

- Course, Attendance, Reference Book
- What do we study in this course?
- Why should this be studied?
- What is “Computer Organization”
- What is “ Computer Architecture”
- Comparison between computer organization and computer architecture

BBT2202/CNS2202 Course

- 67% Attendance is Mandatory
- Manual & **online attendance System**
- 2 assignments + 2 CATs + 1 Presentation
 - **Any kind of copy/plagiarism lead to `F' Grade**

BBT2202/CNS2202 Course

Textbook

- ❑ Computer Architecture: A Quantitative Approach, John L. Hennessy and David A. Patterson, Morgan Kaufmann, 6th ed. (2019)
- ❑ Stalling W., Computer Organization & Architecture, 7th Edition, Pearson Education India, 2010

Reference Book

- ❑ Hennessy J. L., Patterson D., Computer Organization And Design: The Hardware/Software Interface, Morgan Kaufmann, 2008
- ❑ Fundamentals of Computer Organization and Architecture, Mostafa Abd-El-Barr and Hesham El-Rewini, Wiley-Interscience, 1st ed. (2005), ISBN-10: 0471467413 ISBN-13: 978-0471467410

BBT2202/CNS2202 Course

- Course enrollment key: (*to be shared on WhatsApp*)
- Mark distribution
 - 10% : Assignments
 - 20% : CAT
 - 10% presentation
 - 60% : End Semester exam
- Class Timing & Venue
 - Theory class: **Virtual** (Mon: 3:15pm to 5:15pm)
 - Lab class: **Kindaruma Lab** (Wed: 1:15pm to 3:15pm)

BBT2202/CNS2202 Course Objectives

To learn -

- How computers work, basic principles
- How to analyze their performance (or how not to!)
- How computers are designed and built
- Issues affecting modern processors (caches, pipelines, etc.)

BBT2202/CNS2202 Course Motivation

This knowledge will be useful if you need to:

- Design/build a new computer
 - rare opportunity
- Design/build a new version of a computer
- Improve software performance
- Purchase a computer
- Provide a solution with an embedded computer

Course Prerequisite

- Require Knowledge of
 - Software Engineering
 - Data Structure and C/C++ Programming
 - Basic high school Standard Probability and Mathematics

Example1:

- ❑ Say you are in a company that manufactures, design and builds all low-level details of the car like how different functions of the car shall be operated and the flow of commands and instructions, this part of the manufacturing come under **architecture (abstract, programmers view)**.
- ❑ While the manufacturing stage that involves making the parts piece by piece and connecting together the different components of that car by keeping the basic design in mind comes under **organization (physical and visible)**.

What is “Computer Organization”

- Is the implementation of what is specified by the computer architecture .
- It deals with how operational attributes are linked together to meet the requirements specified by computer architecture.
- Some organizational attributes are hardware details and control signals

Input-Output Organization

Peripheral Devices

I/O Subsystem

- Provides an efficient mode of communication between the central system and the outside environment
- Input or Output devices attached to the computer

Input-Output Interface

- A conversion of signal values may be required

Input/output Devices

- Input/output devices are required for users to communicate with the computer.
- In simple terms, input devices bring information INTO the computer and output devices bring information OUT of a computer system.
- These input/output devices are also known as peripherals.

Input Devices are:

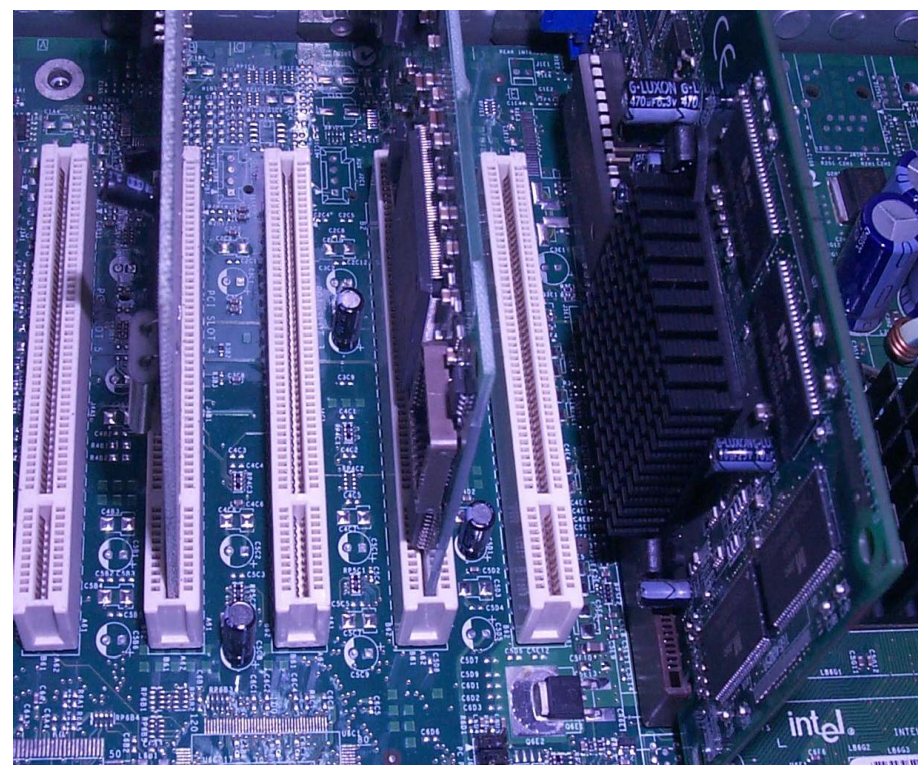
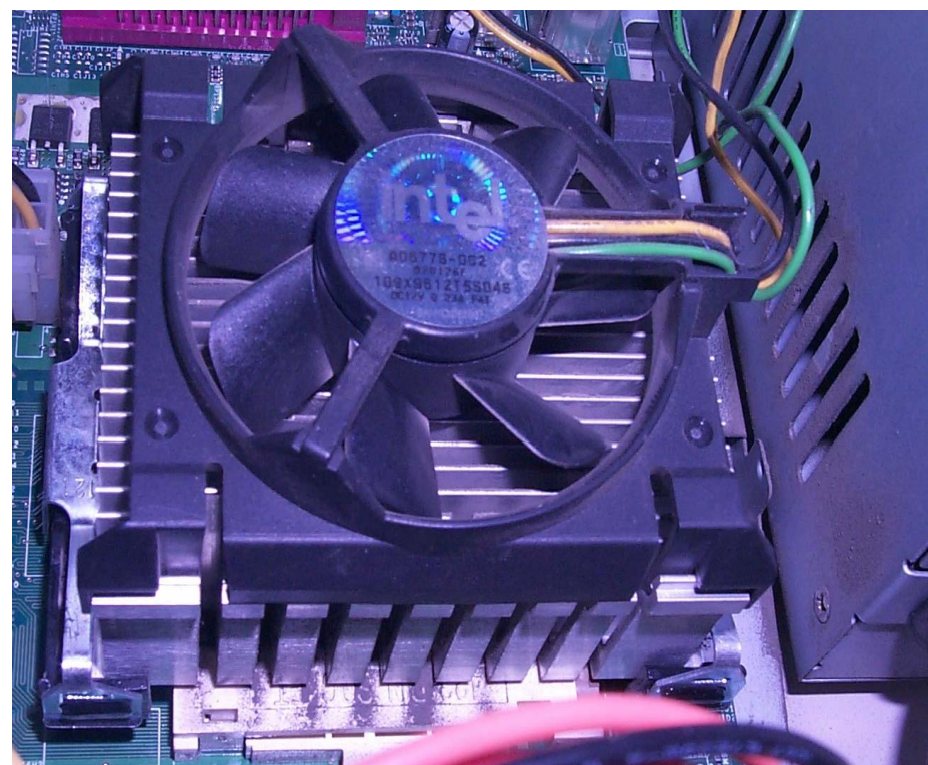
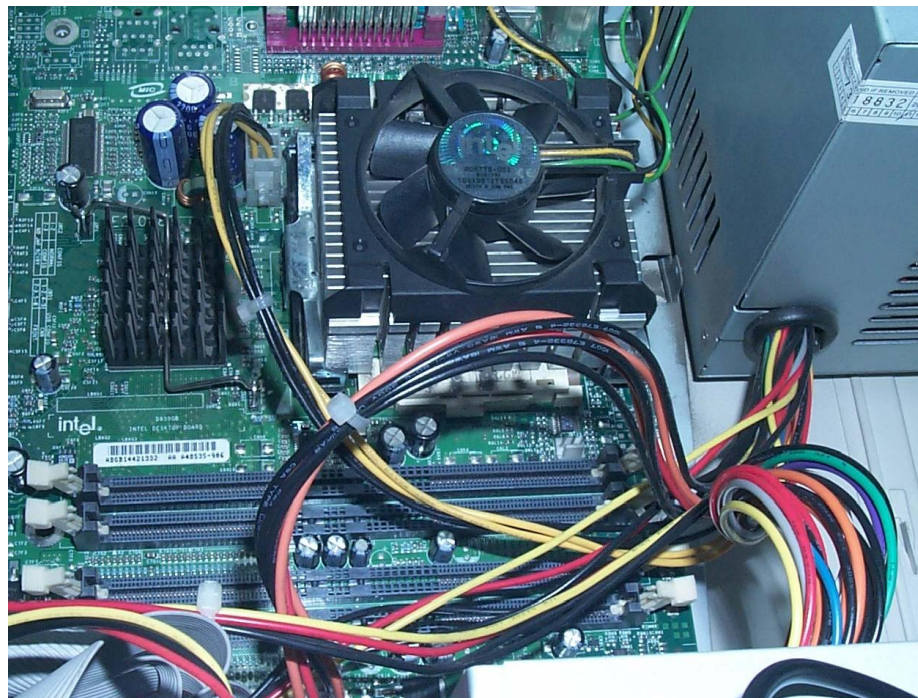
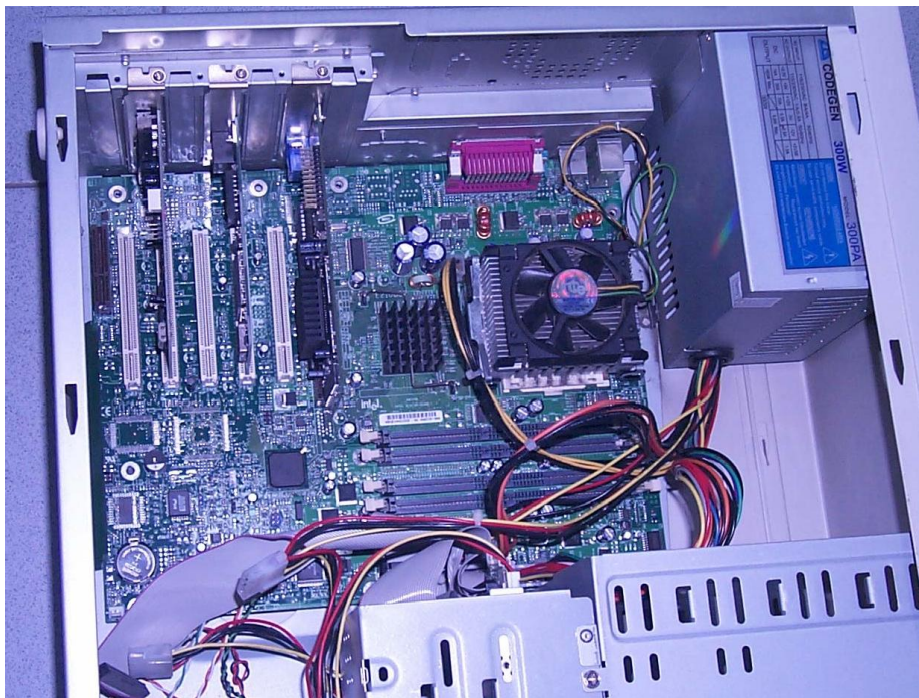
- Keyboard
- Mouse
- Joystick
- Scanner
- Microphone
- Webcam
- Graphic tablet
- Barcode scanner
- Biometric devices
- Touch Screen

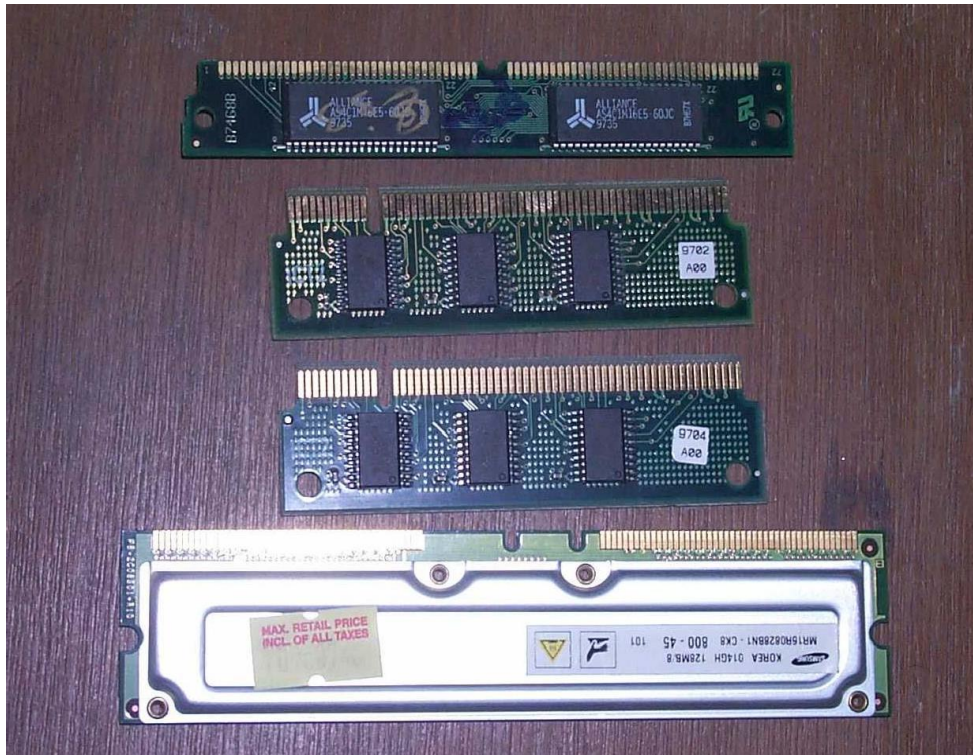
Output devices are

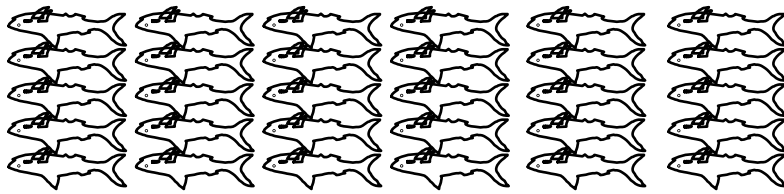
- Printers
- Plotters
- Monitor
- LCD
- Speakers
- Projectors
- Braille display
- Haptic feedback devices (game controllers)

CPU organization

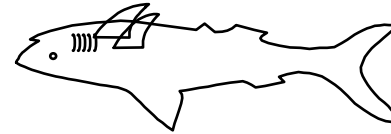
- CPU controls the Computer
- The CPU will fetch, decode and execute instructions.
- The CPU has three internal sections: register section, ALU and Control Unit







Massively Parallel Processors

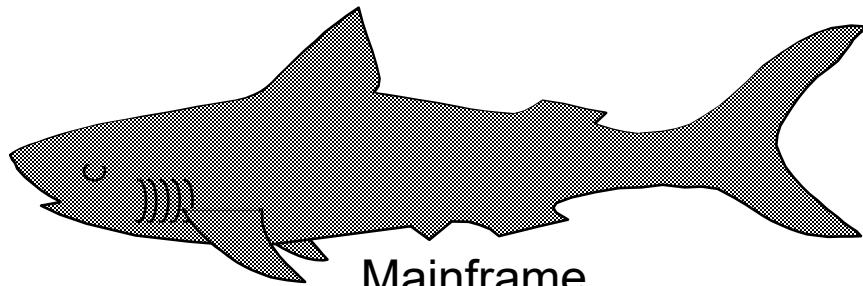


Mini-
supercomputer

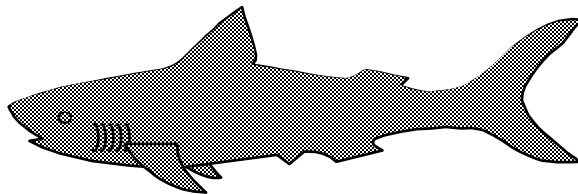


Mini-
computer

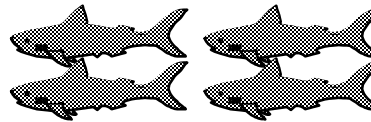
Computer Food Chain



Mainframe



Supercomputer



Server



Work-
station

PC

Now who is eating whom?

What is “Computer Architecture”

- Computer Architecture deals with giving operational attributes of the computer or Processor to be specific.

It deals with details like

1. Physical memory, ISA (Instruction Set Architecture) of the processor
2. The number of bits used to represent the data types
3. Input Output mechanism
4. Technique for addressing memories.

Abstraction

- Delving into the depths reveals more information
- An abstraction omits unneeded detail, helps us cope with complexity

Software Abstraction

```
int sum(int x, int y)
{
    int t = x+y;
    return t;
}
```

C/C++ code

```
sum:
    pushl %ebp
    movl %esp,%ebp
    movl 12(%ebp),%eax
    addl 8(%ebp),%eax
    movl %ebp,%esp
    popl %ebp
    ret
```

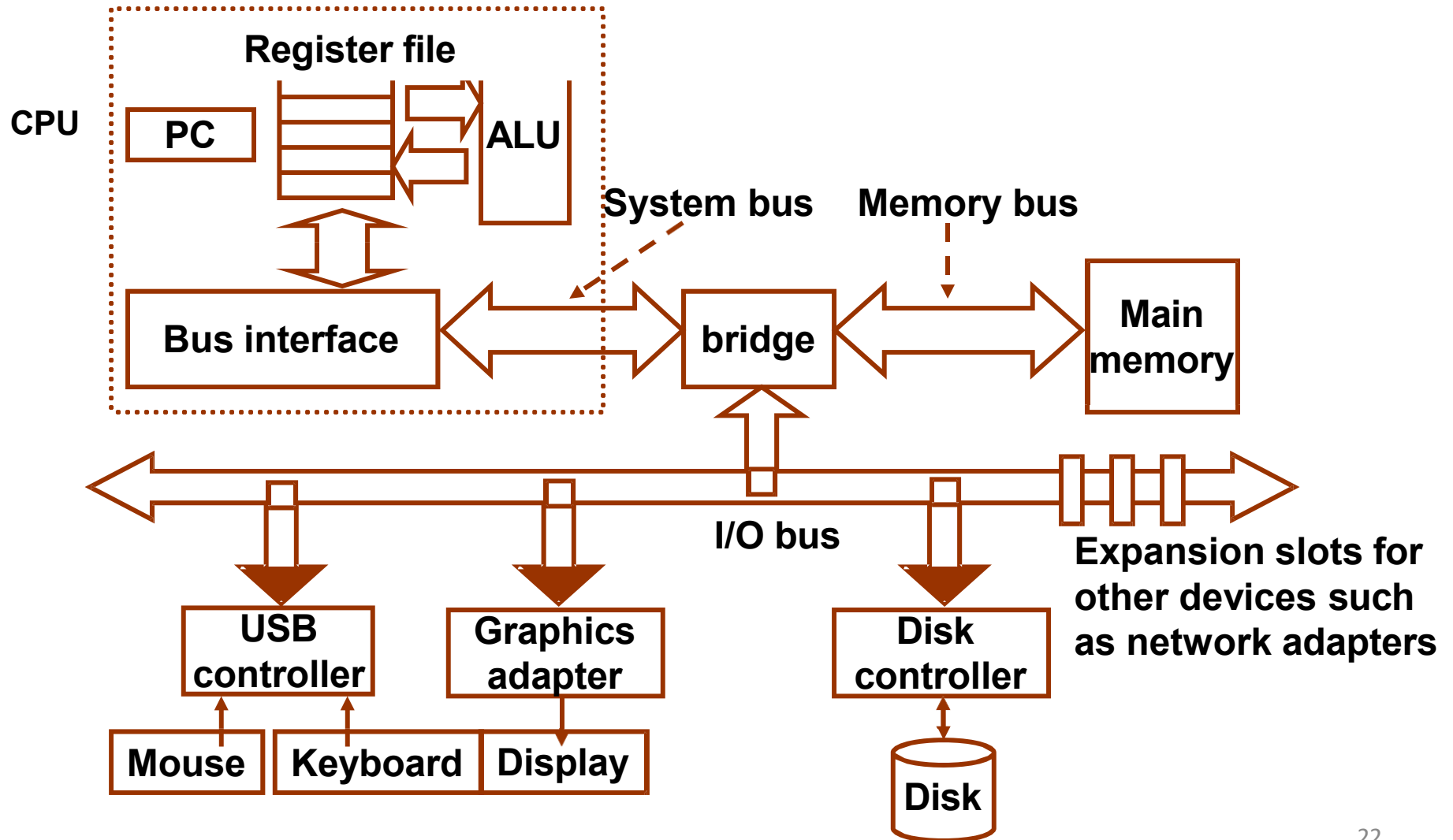
Assembly
code

0x401040 <sum>:

Machine
code

0x55
0x89
0xe5
0x8b
0x45
0x0c
0x03
0x45
0x08
0x89
0xec
0x5d
0xc3

Hardware Abstraction



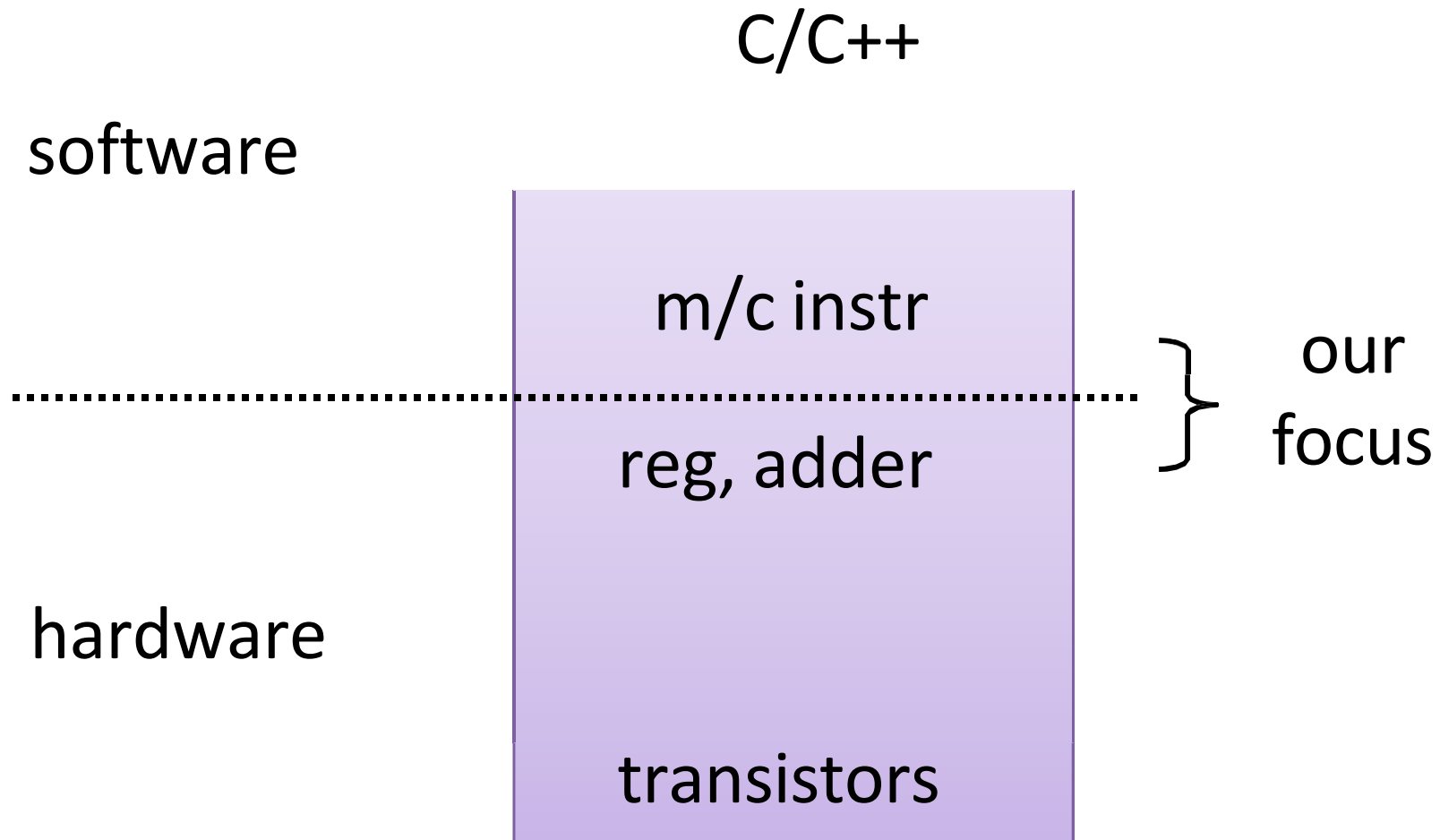
Register Section

- Includes collection of registers and a bus.
- Processor's instruction set architecture are found in this section.
- Non accessible registers by the programmer. These are to be used for registers to latch the address being accessed and a temp storage register.
- They include flag registers

Arithmetic/Logic Unit (ALU)

- Performs most arithmetic and logical operations.
- Retrieves and stores its information with the register section of the CPU.

Hardware/software interface

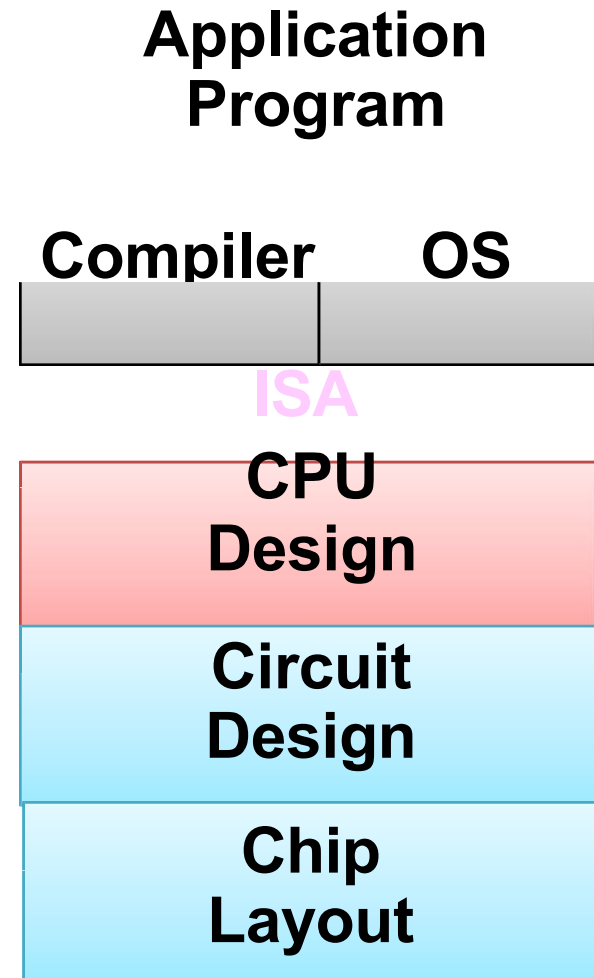


Architecture levels

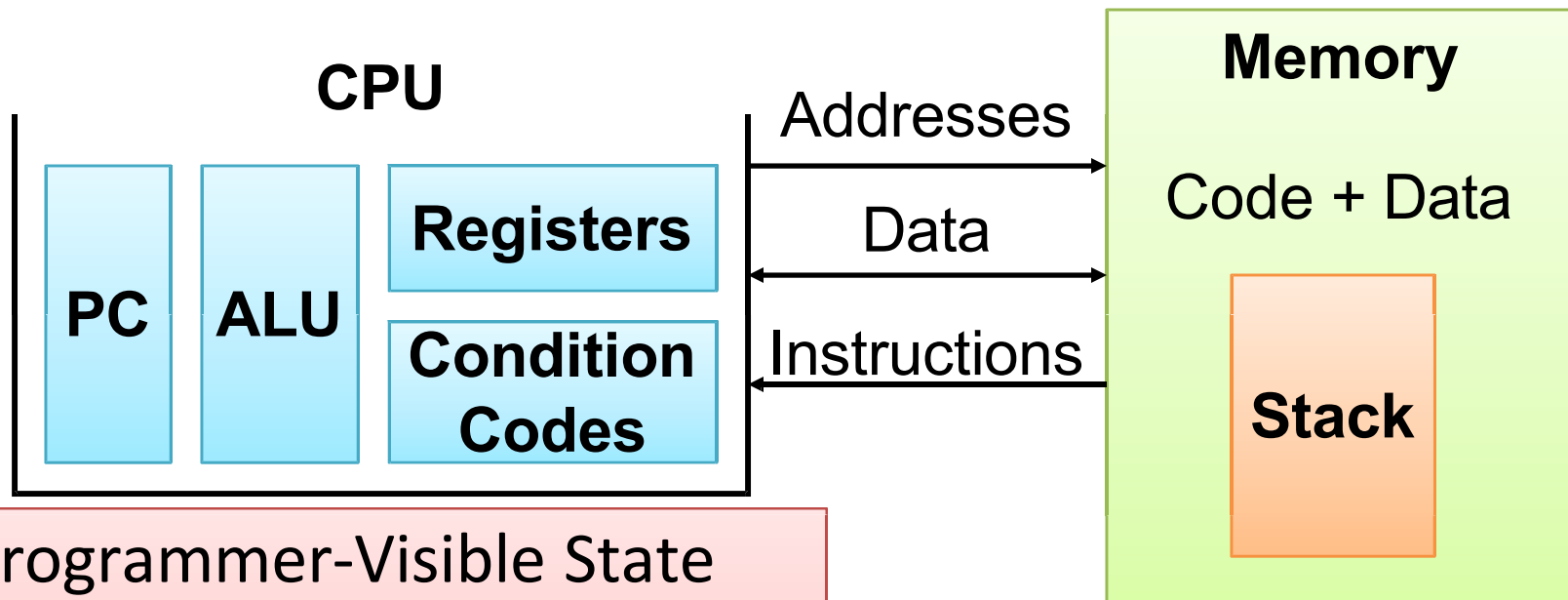
- Instruction set architecture (ISA)
 - Lowest level visible to a programmer
 - Operation (add/sub/mul/shift)
- Micro architecture
 - Fills the gap between instructions and logic modules
 - Operation Vs Micro Operation

Instruction Set Architecture

- Assembly Language View
 - Processor state (memory)
 - Instruction set and encoding
- Layer of Abstraction
 - Above: how to program machine - HLL, OS
 - Below: what needs to be built
 - tricks to make it run fast



The Abstract Machine



- Programmer-Visible State
 - PC Program Counter
 - Register File
 - Heavily used data
 - Condition Codes/flags

□ Memory

- Byte array
- Code + data
- stack

Some realities

- *You've got to know assembly*
 - Chances are, you'll never write programs in assembly, but understanding assembly key to machine-level execution model
- *Memory Matters*
 - Memory is not unbounded
 - Memory performance is not uniform
- *There's more to performance than asymptotic complexity*

Embedded computers

- Processor as an intelligent electronic component **rather than** Processor as a computing engine
- Real time operation
- Requires hardware-software co-design
- Highly customized

Why different processors?

- What is the difference between processors used in desk-tops, laptops, mobile phones, washing machines etc.?
 - Performance / speed
 - Power consumption
 - Cost
 - General purpose / special purpose
 - Domain Specific: Network, DSP, Image. Crypto

Comparison: Computer organization & Computer Architecture

S.No	Computer Architecture	Computer Organization
1	Deals with the way hardware components are connected to form a computer system	Deals with the structure and behavior of computer systems as seen by the use.
2	Acts as an interface between hardware and software	Deals with the components of a connection in a system
3	Helps us understand the functionalities of the computer system	Tells us how exactly all the units in the system are arranged and interconnected.
4	A programmer can view architecture in terms of instructions, addressing modes and registers	Organization is the physical form that helps us understand the architecture
5	Deals with high level design	Deals with low-level design
6	Architecture involves logic components (instruction sets, data type and cache optimization)	Organization involves physical components (circuit design and peripherals, and signals)
7	How to do (instruction set)	What to do (implementation of architecture)